

Ducks Unlimited Chapters

ETL Schedule

Introduction and Ask

- Extract university chapters from the Ducks Unlimited ArcGIS Feature Service
- Filter to California
- Load into Postgres, idempotently
- Run on a daily schedule, unattended
- Package as a container, deploy to Google Cloud as code

Software and Services

- Python
- Cloud Run
- Cloud SQL for PostgreSQL
- Cloud Scheduler
- Artifact Registry
- Secret Manager
- Docker & docker compose
- Terraform (IaC)

Python Programs

Layer	Module	Responsibility
Extract	extract.py	Query, Retry, Pagination
Transform	transform.py	Map, filter by State, Retry
Load	load.py	Upsert, Idempotent
Config	config.py	Environment access
Orchestration	main.py	All program wired

Decisions

1. Runs to completion and exits. A Service must stay resident and bills for idle time the wrong shape for batch work.
2. A single value to rotate. The same code path serves local TCP and the Cloud SQL unix socket.
3. Anything on a schedule eventually runs twice. ON CONFLICT DO UPDATE makes a repeat run harmless.
4. A short-lived token, pinned to one repository no long-lived JSON key sitting in a GitHub secret.

Next Steps

Naming conventions to be improved

Alerting on a non-zero job exit

Historical tracking slowly-changing dimension (SCD)

Backup Server

Remote Terraform state, versioned GCS bucket